

University of Colombo School of Computing

SCS 3312 : Research Methods

D2: Data Collection Design & Pilot Report

**“An Investigation into Student Behavior During Free Periods
Between Scheduled Lectures at UCSC”**

Student Names and Indexes

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Abstract

This report presents the quantitative instrument design and pilot testing phase of a study investigating how undergraduate students at the University of Colombo School of Computing utilise their free periods between scheduled lectures. Building on the qualitative themes identified in the D1 Fieldwork Insights Report, this submission translates nine key behavioural patterns into measurable survey variables, organised across five sections covering participant background, campus space usage and avoidance, factors influencing location choice, activities during free periods, and satisfaction with campus facilities.

A pilot study was conducted with ten UCSC undergraduate students, two from each of the five field sites, using printed survey forms administered at each location. The pilot identified twelve issues across five categories — ambiguous question wording, instructional and response format ambiguity, missing response options, potential bias and sensitivity issues, and survey length and flow — each of which was addressed through a corresponding revision to the instrument.

The finalised survey instrument, incorporating all twelve revisions, is presented in full at the end of this report. It consists of five sections and one optional open-ended question, with an estimated completion time of five to seven minutes, and is ready for full-scale quantitative data collection in the D3 phase.

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List of Acronyms

UCSC	—	University of Colombo School of Computing
LMS	—	Learning Management System

1 Introduction

The D1 phase used observation and short interviews at five locations — the library, canteen, dewale, common room and open area — to study how students spend free time between lectures. D2 builds on these findings by turning them into a measurable survey. The goal is to confirm whether the behavioural patterns identified during D1 hold true across the broader student population, not just the individuals who were directly observed.

This report covers: turning D1 themes into measurable variables (Chapter 2), the survey design (Chapter 3), the pilot test (Chapter 4), the revisions made from pilot feedback (Chapter 5), the final survey (Chapter 6), a reflection on the pilot process (Chapter 7), the full data collection plan (Chapter 8), and a summary (Chapter 9).

Throughout, the goal has been consistency — every survey question links back to a D1 finding, every pilot issue is recorded and addressed, and the final instrument functions as one connected tool rather than a collection of separate parts.

2 Research Objectives and Variable Logic

2.1 Research Objectives

The quantitative phase of this research is based on the key themes identified during the D1 report:

- To identify which location is most frequently used by students during free periods, among the given five locations.
- To identify the main activities students engage in during free periods and measure the relative frequency of each.
- To quantify the prevalence of passive phone browsing as a default free-period activity.
- To measure the extent to which the next lecture hall affects students' decisions about choosing a location.
- To assess student satisfaction with existing campus facilities.
- To identify the relationship between the length of a free period and the activity students engage in.

2.2 From D1 Themes to Survey Design

The themes identified in the D1 fieldwork formed the basis for the survey developed in this report. Patterns such as choosing locations based on the activity planned, the effect of proximity and free-period length on location choice, the prevalence of passive phone use, and the lack of adequate rest space were each turned into a measurable variable. The identified variables are divided into two types as follows.

- **Nominal** (categorical, unordered — no inherent ranking):
 - Single choice (e.g. typical free-period duration)
 - Multi-select (e.g. location avoidance reasons)
 - Binary (yes/no) (e.g. perceived value of phone browsing)
- **Ordinal** (categorical, with an ordering):

- Likert scales of 1–5 (e.g. influence of free-period length on location choice, environmental influence on location choice)
- Ranked choice (e.g. top 3 activities)

The full mapping of each D1 theme to its survey question, variable, type, and scale is given section-by-section in Chapter 3.

3 Survey Instrument Design

3.1 Instrument Overview

The full survey consists of five main sections, each with sub-questions. The instrument uses a combination of single-choice questions, multi-select questions, and Likert scale items to capture both categorical and scaled quantitative data.

Section	Title
A	Background information
B	Space usage and avoidance
C	Factors influencing location choice
D	Activities during free periods
E	Satisfaction with campus facilities

Table 3.1: Survey sections and their corresponding titles

For the purpose of this survey,

“a free period refers to a gap between two timetabled sessions within the academic day. At UCSC, timetabled sessions run during the morning slot (8.00 am – 12.00 pm) and the afternoon slot (1.00 pm – 5.00 pm). The lunch break (12.00 pm – 1.00 pm) and any time before your first session or after your final session of the day are not considered free periods for this study.”

3.2 Section A: Participant Background

No.	Variable	Link with D1	Question	Type	Scale / Options
A1	Year of Study	Year of study may be a significant factor on space preferences	“What is your year of study?”	Single choice	1st / 2nd / 3rd / 4th
A2	Typical free period duration	D1 interviews severely showed that free period duration highly affects both location choice and activity type	“What is the typical duration of your free periods?”	Single choice	<30 min / 30–60 min / 1–2 hrs / >2 hrs
A3	No. of free periods per week	Frequency of free periods affects how regularly a student engages in those locations	“How many free periods do you typically have per week?”	Single choice	0–1 / 2–3 / 4–5 / 6 or more

Table 3.2: Survey instrument — Section A (Participant background)

3.3 Section B: Space Usage and Avoidance

No.	Variable	Link with D1	Question	Type	Scale / Options
B1	Locations visited, ranked by frequency	D1 observations showed that each campus space has a dominant user group and purpose, and that many students use multiple spaces depending on activity and availability	“Which campus locations do you visit more frequently during free periods?” Rank the top 3 locations	Ranked / Multi-select	Library / Canteen / Dewale / Common Room / Open Area
B2	Whether any campus location is actively avoided	D1 interviews showed a significant number of students actively avoid at least one campus space	“Are there any campus spaces you actively avoid during free periods?”	Binary	Yes / No
B3	Avoided location(s) and reason(s) per location	D1 fieldwork identified specific environmental and situational reasons for avoidance of particular locations	“If yes, which campus location(s) do you avoid during free periods, and why?”	Matrix (Check-box grid)	Rows: locations (Library / Canteen / Dewale / Common Room / Open Area) Columns: Reasons (Too noisy / Smell / Crowded / Far / No seating / No Wi-Fi or outlets / Uncomfortable atmosphere / Other)

Table 3.3: Survey instrument — Section B (Space usage and avoidance)

3.4 Section C: Factors Influencing Location Choice

No.	Variable	Link with D1	Statement	Type	Scale / Options
C1	Influence of proximity to next lecture	D1 interviews showed that most students select a particular location based on how close it is to their next lecture hall	“I choose my free period location based on how close it is to my next lecture hall.”	Likert	1–5 (1=Never, 5=Always)
C2	Influence of free period length on location choice	D1 interviews revealed a clear behavioural pattern — students with short gaps tended to stay nearby, while those with longer gaps were more likely to go to the library	“The length of my free period affects where I choose to spend it.”	Likert	1–5 (1=Strongly disagree, 5=Strongly agree)
C3	Social motivation for location choice	According to D1 interviews, friends’ presence highly affects in selecting location	“I choose locations where I expect to find friends or classmates.”	Likert	1–5 (1=Strongly disagree, 5=Strongly agree)
C4	Activity-location fit preference	D1 observations showed that students self-sort by activity across campus spaces	“I prefer locations that support the specific activity I plan to do (e.g., studying, resting).”	Likert	1–5 (1=Strongly disagree, 5=Strongly agree)
C5	Environmental influence on choice	D1 open area observations showed that student presence dropped significantly during hot or uncomfortable weather	“Environmental conditions (e.g., heat, noise) influence my choice of location.”	Likert	1–5 (1=Strongly disagree, 5=Strongly agree)

Table 3.4: Survey instrument — Section C (Factors influencing location choice)

3.5 Section D: Activities During Free Periods

No.	Variable	Link with D1	Question	Type	Scale / Options
D1	Primary Activities (top 3 ranked)	D1 observations across all five locations identified a consistent range of recurring activities	“What do you most commonly do during free periods?” Rank the top 3 activities	Ranked / Multi-select	Study/review notes / Eat or drink / Social conversation / Browse phone / Sleep-rest / Recreational activities / Club or society meetings
D2	Frequency of Phone Browsing	D1 observations consistently showed phone browsing as the default fallback activity across multiple locations	“How often do you browse your phone when you have nothing specific to do?”	Likert	1–5 (1=Never, 5=Almost Always)
D3	Perceived value of phone browsing	D1 interview responses suggested that students held mixed views on their own phone use	“Do you consider phone browsing during free periods to be a productive use of time?”	Binary	Yes / No
D4	Solo vs group tendency	Observations showed a clear distinction between students who consistently occupied spaces alone and those who consistently sought group settings	“Do you typically spend free periods alone or with others?”	Single choice	Always alone / Mostly alone / Mostly with others / Always with others

No.	Variable	Link with D1	Question	Type	Scale / Options
D5	Influence of friend availability on planned activity	D1 interviews showed that students who were alone often engaged in academic or passive activities, while the same students shifted toward social or leisure activities when friends were present	“When friends are available during a free period, do you tend to change the activity you had planned to do?”	Likert	1–5 (1=Never, 5=Almost Always)

Table 3.5: Survey instrument — Section D (Activities during free periods)

3.6 Section E: Satisfaction with Campus Facilities

No.	Variable	Link with D1	Statement	Type	Scale / Options
E1	General satisfaction with campus spaces	D1 interviews revealed mixed levels of satisfaction with available campus spaces	“Overall, I am satisfied with the campus spaces available to me during free periods.”	Likert	1–5 (1=Strongly Disagree, 5=Strongly Agree)
E2	Sufficiency of seating	Observations at the canteen and open area during peak hours showed that seating was frequently insufficient	“There is sufficient seating available during peak free period times.”	Likert	1–5 (1=Strongly Disagree, 5=Strongly Agree)
E3	Adequacy of campus spaces for the range of student activities	In D1 observations, some students said that the space does not accommodate enough free-time activities they want to perform	“Campus spaces adequately support the range of activities I want to do during free periods.”	Likert	1–5 (1=Strongly Disagree, 5=Strongly Agree)

Table 3.6: Survey instrument — Section E (Satisfaction with campus facilities)

3.7 Open-Ended Question

One optional open-ended question is included at the end of the survey to allow participants to provide insights not captured by structured items:

“Is there anything else you would like to share about how you use campus spaces during free periods, or any improvements you feel would benefit students, that is not mentioned above?”

4 Pilot Study

4.1 Pilot Study Design

A pilot test was conducted before the main data collection to identify issues related to question clarity, response options, and the overall flow of the survey. The pilot was designed to closely reflect the conditions of the actual data collection process.

Pilot parameter	Detail
Sample Size	10 UCSC undergraduate students — 2 from each of the five field sites (Library, Canteen, Dewale, Common Room, Open Area)
Recruitment method	Convenience sampling, stratified by location — students already present at each field site during free period hours were approached and invited to participate
Administration Mode	Printed survey forms were distributed among the participants and asked them to fill it
Administration setting	Each student completed the survey at their own location without being moved elsewhere
Duration	Conducted across two days during free period hours
Debrief	After completing the survey, each student was asked a few short verbal questions about whether any questions were confusing, unclear, or missing relevant options

Table 4.1: Pilot study design parameters and corresponding details

4.2 Pilot Findings: Issues Identified

The following issues are identified by examining the participants' responses and through the debrief conducted with the participants.

4.2.1 Ambiguous Question Wording

Question	Issue Identified	How Identified
A2 (free period duration)	Students were unsure whether to report their most common or average duration	Responses varied widely; three students circled two options. In the debrief, participants mentioned that they were confused what to select
A3 (free periods per week)	Although the survey definition established the general time window for free periods, the question did not clarify whether to count only timetable-based gaps, unplanned gaps from unexpected cancellations, or gaps created by choosing not to attend a session	Three respondents gave noticeably higher counts than their regular timetable would suggest, and later in the debrief they mentioned that they had counted lectures they usually do not attend as free periods
C3 (“I choose locations where I expect to find friends...”)	“Expect to find” was unclear — unclear whether it meant deliberately choosing a location because friends will be there, or simply anticipating that friends are usually present regardless of intent	Verbal debrief revealed many respondents interpreted it as a deliberate social choice, but some as a passive expectation

Table 4.2: Pilot findings — ambiguous question wording: issues identified and how identified

4.2.2 Instructional / Response Format Ambiguity

Question	Issue Identified	How Identified
B1 (Rank top 3 locations) D1 (Rank top 3 activities)	Ranking instruction did not specify the response mechanism	Some students numbered 1–3; others circled three items as a set
B3 (Avoided locations and reasons — checkbox grid)	The matrix table instructions did not explain how to fill in the grid (students were not told whether to tick one reason or multiple)	Some students ticked all the reasons for a particular location, while two members ticked one reason per location, even though they had multiple reasons as told in the debrief, because they were not sure about the correct format. Also one student left the table empty, even though they answered ‘yes’ in the previous question

Table 4.3: Pilot findings — instructional and response format ambiguity: issues identified and how identified

4.2.3 Missing Response Options

Question	Missing Options Identified	How Identified
D1 (Activities)	“Attend an online meeting or class” and “Attend to personal admin tasks (e.g. checking marks or results in the LMS / Submitting an assignment, going to the administration counter to sort out something)” are not given as options in the activities list	One respondent wrote attending an online meeting as an activity performed during free period in the ‘other’ section, while another participant mentioned submitting assignments as an activity
B3 (Avoided locations and reasons for avoidance)	“Lack of Wi-Fi or power outlets” is not given as a reason in the checkbox grid	One respondent wrote it as a reason for not selecting Dewale, in the ‘other’ section

Table 4.4: Pilot findings — missing response options: issues identified and how identified

4.2.4 Potential Bias and Sensitivity Issues

Question	Potential Bias
D3 (“Do you consider phone browsing to be a productive use of time?”)	The word “productive” carries a social stigma that pushes respondents toward answering “No” to avoid appearing lazy, and the binary Yes/No format makes this worse by removing any middle-ground option — respondents are forced to make an all-or-nothing judgment about their own behaviour
C1 (“I choose my free period location based on how close it is to my next lecture hall”)	Leading statement — ‘I choose my free period location based on. . . ’ presupposes the answer rather than asking neutrally
D1 (‘Sleep / rest’ activity option)	Students may underreport sleeping as a ranked activity due to social embarrassment, despite D1 fieldwork observing it as common and normalised in the Common Room and Library
D4 (“Do you typically spend your free period alone or with friends?”)	Answers are given as “Always alone / Mostly alone / Mostly with others / Always with others”. Students who usually spend free periods alone may not answer honestly, as selecting “always alone” or “mostly alone” can feel embarrassing — it may come across as admitting they have no friends rather than simply describing how they prefer to spend their time

Table 4.5: Pilot findings — potential bias and sensitivity issues: issues identified and action taken

4.2.5 Survey Length and Flow

In the debrief, two pilot respondents mentioned that the survey felt slightly long and suggested the open-ended question should be clearly marked as optional.

5 Survey Revisions

Based on the revised pilot findings above, the following changes were made to the survey instrument before finalising it for full-scale data collection.

Rev. No.	Question Affected	Revision Made
R1	A2 (free period duration)	Added instruction: “Select the option that best describes your most common free period length”
R2	A3 (free periods per week)	Definition of free period updated in the final instrument to explicitly state that only timetable-based planned gaps are counted, excluding unexpected cancellations and voluntary non-attendance
R3	C3 (social location choice)	Reworded the statement to: “I choose locations where I am likely to find friends or classmates”
R4	B1 (Rank top 3 locations) and D1 (Rank top 3 activities)	A small box was added in front of each option for respondents to write their rank (1, 2, or 3). The instruction was updated to: “Write 1, 2, or 3 in the box next to your choices (1 = most frequent / most common). Leave all other boxes empty.” A worked example was also added
R5	B3 (Avoided locations and reasons — checkbox grid)	Added explicit instruction above the grid: “For each location you avoid, tick all reasons that apply in that row. Leave a row blank if you do not avoid that location.” A worked example row was also added
R6	D1 (Activities)	Added the following two options to the activity list: (i) “Attend an online meeting or class”; (ii) “Personal admin tasks (e.g., checking marks, submitting assignments, visiting admin counter)”
R7	B3 (location avoidance reasons)	Added “Lack of Wi-Fi or power outlets” to the B3 checkbox grid columns
R8	D3 (Phone use)	Reworded the question to: “How do you feel about phone browsing during your free periods?” with options: Fulfilling / Neutral / Unfulfilling / Depends

Rev. No.	Question Affected	Revision Made
R9	C1 (Proximity)	Reframed as a neutral frequency question: “How often does the location of your next lecture influence where you spend your free period?”
R10	D1 (“Sleep / rest” option)	Reworded “Sleep” into “rest / take a short break” to reduce stigma while retaining the same behavioural meaning
R11	D4 (Solo vs group tendency)	Reframed as a Likert frequency question: “How often do you spend your free periods with friends or classmates?” (1 = Never ... 5 = Almost always)
R12	Open-ended question (final item)	Added bold “(Optional)” label and a clarifying note above the open-ended question

Table 5.1: Survey revisions made following pilot study findings

6 Final Survey Instrument

The finalised survey instrument, incorporating all revisions from the pilot phase, is presented below in full. It consists of five sections and one optional open-ended question, with an estimated completion time of 5–7 minutes.

SURVEY: Student Behavior During Free Periods at UCSC

This survey is part of a research study on how students use campus spaces during free periods. Your participation is voluntary and anonymous. It should take approximately 5–7 minutes to complete

Definition — Free Period: For the purpose of this survey, a free period refers to a planned gap between two timetabled sessions within the academic day — that is, a block of time that appears as empty on your regular timetable. Gaps created by unexpected session cancellations or sessions you choose not to attend are not considered free periods for this study. At UCSC, timetabled sessions run during the morning slot (8.00 am – 12.00 pm) and the afternoon slot (1.00 pm – 5.00 pm). The lunch break (12.00 pm – 1.00 pm) and any time before your first session or after your final session of the day are also excluded.

Section A: Background Information

A1. What is your year of study?

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year

A2. What is the typical duration of your free periods?

(Select the option that best describes your most common free period length.)

- Less than 30 minutes
- 30–60 minutes
- 1–2 hours

- More than 2 hours

A3. How many free periods do you typically have per week?

- 0–1
- 2–3
- 4–5
- 6 or more

Section B: Space Usage and Avoidance

B1. Which campus locations do you visit during free periods? Rank the locations you use in order of frequency.

(Write 1, 2, or 3 in the box next to your choices (1 = most frequent). Leave all other boxes empty.)

Example:

1	Study or re-view notes		Eat or drink	2	Browse phone
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- ☐ Library
- ☐ Canteen
- ☐ Dewale
- ☐ Common Room
- ☐ Open Area

B2. Are there any campus spaces you actively avoid during free periods?

(If No, skip to Section C)

- Yes
- No

B3. If Yes — which campus location(s) do you avoid, and why?

(Tick all reasons that apply in each row. Leave a row blank if you do not avoid that location.)

Example: If you avoid the Canteen because it is too noisy and too crowded, tick both of those boxes in the Canteen row.

	Too noisy	Unpleasant smell	Too crowded	Too far	No seating available	No Wi-Fi / power outlets	Uncomfortable atmosphere	Other (please mention)
Library								
Canteen								
Dewale								
Common Room								
Open Area								

Section C: Factors Influencing Location Choice

Indicate how much you agree or disagree with each statement by ticking the correct column for particular statement. (1 = Strongly Disagree, 5 = Strongly Agree)

Statement	1	2	3	4	5
C1. How often does the location of your next lecture influence where you spend your free period? (1 = Never, 5 = Always)					
C2. The length of my free period affects where I choose to spend it.					
C3. I choose locations where I am likely to find friends or classmates.					
C4. I prefer locations that match the activity I plan to do (e.g., studying, resting).					
C5. Environmental conditions (e.g., heat, noise) influence my choice of location during free periods.					

Section D: Activities During Free Periods

(This section asks about what you typically do during free periods and how you use your time.)

D1. What do you most commonly do during free periods? Rank your top 3 activities. (Write 1, 2, or 3 inside the box next to your choices (1 = most common). Leave all

other boxes empty.)

Example:

1	<i>Study or re-view notes</i>		<i>Eat or drink</i>	2	<i>Browse phone</i>
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- ☐ Study or review lecture notes
- ☐ Eat or drink
- ☐ Have social conversations with friends
- ☐ Browse phone / scroll social media
- ☐ Sleep or rest / take a break
- ☐ Recreational activities (games, carom, cards, etc.)
- ☐ Attend club or society meetings
- ☐ Personal admin tasks (timetable, forms, LMS, admin counter, etc.)
- ☐ Attend an online meeting or class
- ☐ Other (please mention) : _____

D2. How often do you browse your phone during free periods when you have nothing specific to do?

(1 = *Never*, 3 = *sometimes*, 5 = *Almost always*)

- 1
- 2
- 3
- 4
- 5

D3. How do you feel about phone browsing during your free periods?

- Fulfilling
- Neutral
- Unfulfilling
- It depends on the situation

D4. How often do you spend your free periods with friends or classmates?

(1 = *Never*, 3 = *Sometimes*, 5 = *Almost always*)

- 1
- 2
- 3

- 4
- 5

D5. When friends are available during a free period, do you tend to change the activity you had planned to do?

(1 = Never, 3 = sometimes, 5 = Almost always)

- 1
- 2
- 3
- 4
- 5

Section E: Satisfaction with Campus Facilities

Indicate how much you agree or disagree with each statement by ticking the correct column for particular statement. (1 = Strongly Disagree, 5 = Strongly Agree)

Statement	1	2	3	4	5
E1. Overall, I am satisfied with the campus spaces available to me during free periods.					
E2. There is sufficient seating available on campus during peak free period times.					
E3. Campus spaces adequately support the range of activities I want to do during free periods.					

Optional Open-Ended Question

(Optional) Is there anything else you would like to share about how you use campus spaces during free periods, or any improvements you feel would benefit students?

Thank you for completing this survey!

7 Critical Analysis

7.1 Value of the Design-Pilot-Revision Cycle

Overall, the process of designing, testing, and improving the questionnaire made it much better than if it had only been written once. Some of the 12 changes, especially clarifying the instructions for the B3 checkbox grid and improving the ranking instructions in B1 and D1, solved major problems that could have resulted in incorrect or unusable data. Other changes, such as rewriting D3 and restructuring C1 and D4, were found only because participants were asked to explain what they found confusing or uncomfortable after completing the survey. This shows that it is not enough to look only at people's answers. It is also important to ask for their feedback to understand any problems with the questionnaire.

The pilot study also helped confirm that the questionnaire was working as expected. Some of the pilot results, such as students who reported longer free periods tended to rank studying as their primary activity and identify the library as their most visited location, matched the findings from the earlier D1 qualitative study that free-period duration shapes both location choice and activity type. Since both methods produced similar results, the team can be more confident that the final questionnaire measures the right topics and provides reliable information.

7.2 Limitations

- The pilot study included only ten students (2 students per each location). They were selected from the same locations used in the D1 observation, so the sample may not represent all students.
- The participants knew they were testing the questionnaire, not taking part in the actual survey. Because of this, they may have read the questions more carefully and pointed out more problems than normal respondents. Some unclear questions may still remain.
- The questionnaire depends on students reporting their own experiences. This can lead to mistakes if they do not remember correctly. Even after improving question D3, some students may still give answers they think are more socially acceptable instead of being completely honest.
- The pilot study was carried out over only two days. Students' behaviour and availability may be different at other times of the semester, especially during

exam periods.

7.3 Ethical Considerations

Before completing the pilot survey, all participants were told about the purpose of the research. Taking part was completely voluntary, and no personal information that could identify them was collected. They were also encouraged to give feedback about the questionnaire itself, not just answer the questions, so that the survey could be improved. No rewards or incentives were given, so participants did not feel pressured to take part or give certain answers. These same ethical principles, including informed consent, voluntary participation, and complete anonymity, will also be followed during the full-scale data collection in D3.

8 Data Collection Plan

The finalised instrument presented in Chapter 6 will be deployed during the full-scale quantitative data collection phase (D3). The parameters below set out how that deployment will be carried out.

Parameter	Plan
Target population	UCSC undergraduate students currently enrolled in courses with scheduled free periods, across 1st, 2nd, 3rd, and 4th years
Target sample size	Minimum 60 respondents, sufficient to support basic subgroup comparisons (e.g. by year of study) alongside overall frequencies and Likert-item means
Sampling method	Stratified convenience sampling, distributed across all five field sites so that no single location dominates the sample
Administration mode	Printed survey forms distributed and completed on paper at each field site
Distribution method	Survey forms handed directly to students present at each of the five campus locations during free period hours, ensuring respondents are students who genuinely occupy those spaces during unstructured time
Data collection period	2nd July 2026 to 10th July 2026
Anonymity	No names, index numbers, or other identifying information will be collected at any point in the survey

Table 8.1: Full-scale quantitative data collection parameters and plan for D3

9 Conclusion and Future Work

9.1 Conclusion

This report explained the complete process of developing a reliable survey, starting from the qualitative findings in D1 and ending with a validated questionnaire that is ready for the main data collection. Based on the themes identified during the observations and interviews — such as students choosing places because they are nearby, spending free time on their phones, following different behaviours in different spaces, and the need for better rest areas — the team created survey questions to measure each of these topics. Instead of assuming the questionnaire was correct, it was tested with real UCSC students.

The pilot study with ten students was very useful. It found important problems, such as unclear instructions for the B3 checkbox grid and unclear ranking instructions in B1 and D1, which could have affected the quality of the data. It also helped identify questions that were confusing or sensitive, allowing the team to improve the wording. All 12 changes made after the pilot study are explained in Chapter 5, and the final 19-question survey is presented in Chapter 6.

As a result, the team now has a well-designed questionnaire that is based on real observations, tested with real students, and improved using their feedback. The data collection plan described in Chapter 8 provides a clear way to collect at least 60 responses for the main study. In the next phase, D3, this questionnaire will be used to collect the final data, and the results will later be compared with the D1 qualitative findings in the D4 final report to give a better understanding of how UCSC students spend their free periods.

9.2 Future Work

D3 — Data Summary and Preliminary Findings: The quantitative data collected during the study will be organised, summarised, and analysed using descriptive statistical methods such as frequencies, percentages, and basic comparisons. Preliminary findings regarding student activity patterns, preferred locations, peak usage periods, and behavioural trends will be identified and compared with the qualitative observations gathered in this phase.

D4 — Final Research Report: The final stage of the study will integrate both qualitative and quantitative findings into a comprehensive analysis of student behaviour during free periods within the UCSC environment. The final report will

critically evaluate behavioural patterns, campus space utilisation, informal learning environments, and student experiences while discussing the strengths, limitations, and implications of the research findings.